

Elastic modulus and stress–strain response of human enamel by nano-indentation

Li Hong He^a, Naoki Fujisawa^b, Michael V. Swain^{a,b,*}

^a*Biomaterials Science Research Unit, Faculty of Dentistry, University of Sydney, Sydney Dental Hospital, Slurry Hills, NSW 2010, Australia*

^b*School of Aerospace, Mechanical & Mechatronic Engineering, University of Sydney, NSW 2006, Australia*

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Abstract

Nano-indentation with a sharp (Berkovich) and two spherical indenters with nominal tip radii of 5 and 20 μm was used to determine the elastic modulus and stress–strain response of human enamel. Indentation tests were made over a wide range of peak loads from 1 to 450 mN in two orthogonal directions, i.e., parallel and perpendicular to enamel prisms. The elastic modulus and hardness (mean contact pressure) versus depth of penetration were determined for the three indenters. From the spherical indentation data, stress–strain curves (H – $\tan \theta$ curve) of enamel were determined in the two orthogonal directions and were found to be different. The elastic modulus showed load dependence for both orientations of the enamel rod structure that depended on the indenter. However, these differences could be normalized upon considering the contact diameter. The indented sample was imaged with an SEM to investigate the near surface damage. In conclusion, prism-sheath structure played an important role in determining the mechanical properties as well as the localized fracture of enamel.

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1. Introduction

Enamel, the outer cover of a tooth, is the hardest and stiffest structure of the body. The functional requirements of teeth are that they be able to bear the range of imposed loads and consequent contact-induced stresses without failure and retain their shapes while doing so [1]. Besides the normal loads, enamel will bear shear forces because of the direct contact with opposite teeth and external objects during mastication. Moreover, unlike other calcified skeletal structures, fracture of dental tissue is not repairable. The ability of enamel to meet such critical requirements has resulted in the structure and mechanical properties of this natural material attracting numerous researchers' interests.

Enamel is composed of inorganic mineral crystals and a small volume of water and protein that holds the inorganic crystals together. The inorganic part of enamel is a form of carbonate hydroxyapatite and the organic part is mainly protein. Though free water and protein comprise only a minor part of mature enamel, they are crucial to its development and are important in understanding its structural organization and physical properties [2]. After maturation, some acidic proteins called enamelines and tuftelins remain inside enamel [3] and act as a “glue” between crystallites [4] and as the cover sheath of the prisms or rods that extend from the dentine enamel junction to the enamel surface. Most free water within enamel is bound within the sheath structure and has an influence on enamel's compressibility, permeability and ionic conductivity [5]. The resulting composite material is much tougher than apatite mineral alone [6]. The mineral and non-mineral components are organized into a complex textured nano-crystallite material that very effectively dissipates forces applied to teeth and protects them from fracture.

*Corresponding author. Biomaterials Science Research Unit, Faculty of Dentistry, University of Sydney, Sydney Dental Hospital, Slurry Hills, NSW 2010, Australia. Tel./fax: +61 02 9351 8375.

E-mail address: mswain@mail.usyd.edu.au (M.V. Swain).

Although the high mineral content accounts for enamel's hardness, it is the arrangement and organization of its inorganic and organic constituents (enamel micro-structure) that modulates the way enamel responds to stress [2]. Enamel structure may be visualized as a number of hierarchical levels of increasing complexity and scale from crystal level to pattern level [2,7]. Among them, the most useful and practical model from a mechanical perspective are the prism rods. Human enamel at this level is composed of a rod and inter-rod sheath structure [8]. The diameter of keyhole-like rods is about 5 μm . They align parallel from enamel–dentin junction to the surface of the tooth and are covered by a thin organic sheath. Recently, an FEA model at this level has been developed to predict the orientation and mineral dependence of the elastic properties of enamel [9].

The well arranged prism micro-structure results in anisotropy of enamel's mechanical properties. During the past half-century, several methods, including both macroscopic and microscopic approaches, were used to determine the mechanical properties such as elastic modulus, strength and toughness of dental tissues [10–20]. Despite such studies, there are virtually no direct measurements of the stress–strain response of enamel that have been reported.

Since the introduction of the spherical indentation method to nano-indentation by Field and Swain [21] and its subsequent verification using the Oliver and Pharr [22] approach, spherical indenters have been used for measuring elastic–plastic and stress–strain properties of some engineering materials [23–28]. In this paper, we extend this approach and develop a new method for measuring the stress–strain relationship of enamel in different directions and compare the results with those obtained using a pointed Berkovich indenter.

2. Theory

In the field of nano-indentation, the Berkovich tip has been widely used to measure the mechanical properties of bulk materials and thin films. This indenter has a triangular shaped tip with the total included angle of 142.3° and half-angle of 65.35°; this tip can only produce a constant equivalent strain value during a test procedure. To measure the stress–strain response of the material, changing strain values are necessary during the course of the test. A spherical tipped indenter becomes a logical choice to meet this need.

2.1. Calculation of stress–strain curve

Determination of the stress–strain response of small volumes of materials has been a problem for a long time in materials science research because macroscale methods such as compression or tension tests are virtually impossible for such small samples. The development of depth sensing indentation technique and spherical indent methods [21] provide a novel approach to this problem.

In his book *Hardness of Metals*, Tabor [29] showed that an equivalence existed between conventional stress–strain curves and that generated by plotting hardness versus contact radius divided by indenter radius (a/R) using a spherical tipped indenter. In the case of an axi-symmetric indenter (sphere, cone or pyramid) in contact with a surface, an appropriate expression for representative strain is the tangent of the complimentary angle between the original specimen and the indenter flank, $\tan \theta$ (see Fig. 1), and contact pressure, H , is the relevant stress. Therefore, to acquire the stress–strain relationship (H – $\tan \theta$ curve), the only data required are the angle θ and contact pressure at a certain indentation depth.

In the nano-indentation field, Oliver and Pharr [30] and Field and Swain [21] have both shown that their methods may apply to spherical indenters. With this method, the elastic contact stiffness, S , can be calculated as dP/dh at the onset of unloading. On the basis of the Sneddon's solution [31] to the contact problem of an elastic half-space loaded with an axis-symmetric punch, S is expressed also as

$$S = \beta \frac{2}{\sqrt{\pi}} E_{\text{eff}} \sqrt{A} \quad (1)$$

in which A is the projected area of contact, and β is a constant associated with the indenter shape. For a spherical indenter, $\beta = 1$. E_{eff} is the reduced or indentation modulus, expressed as

$$\frac{1}{E_{\text{eff}}} = \frac{1 - \nu^2}{E} + \frac{1 - \nu'^2}{E'}, \quad (2)$$

where E , E' and ν , ν' are the Young's moduli and the Poisson's ratios of the specimen and the indenter, respectively. Using fused silica as a standard calibration material, which may be assumed to possess a constant elastic modulus, the radius of contact, a , at a certain force, can be calculated as

$$a = \frac{S}{2E_{\text{eff}}}. \quad (3)$$

The depth of indenter in contact with the material, h_p , can be determined using the Oliver–Pharr method as [22]

$$h_p = h_t - 0.75 \frac{P_t}{S}, \quad (4)$$

where h_t and P_t are the peak displacement and load at the onset of unloading. With the information of the contact

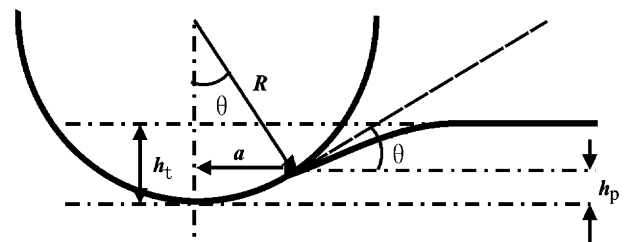


Fig. 1. Schematic diagram of the contact between a spherical indenter and an originally flat specimen.

radius and contact depth of fused silica, the $a-h_p$ curve, which represents the tip shape, may be determined.

From Fig. 1, it is not difficult to show that the contact area of a spherical indenter with a specimen can be described by the h_p-a curve. The tangent angle of this curve, θ , may also be easily calculated from the derivative of the function $h_p(a)$. Moreover, to facilitate calculation, the equation of the curve can be expressed as $a=f(h_p)$ and its derivative as $a'=\tan\beta=f'(h_p)$, in which β is the tangent angle of the curve. In this way, the stress-strain curve of any material can be acquired by collecting P and h_p data alone. For $\theta+\beta=90^\circ$, $\tan\theta$ can be calculated as $\tan\theta=1/\tan\beta=1/f'(h_p)$. (5)

Contact pressure, H , can be calculated as

$$H = \frac{P_t}{A} \quad (6)$$

in which $A=\pi a^2$ and $a=f(h_p)$. With Eqs. (5) and (6), the stress-strain curve ($H-\tan\theta$ curve) may be calculated. Another geometrical relationship in Fig. 1 is

$$a/R = \sin\theta = 1/\sqrt{(\tan^2\theta + 1)}. \quad (7)$$

By combination of Eqs. (3) and (7), an effective indenter radius, R , may be determined. The elastic modulus of the sample may be calculated from Eq. (3) as

$$E_{\text{eff}} = \frac{S}{2a}. \quad (8)$$

With the help of tip shape function $a=f(h_p)$, the elastic modulus of any sample can be measured with spherical tips by using only h_p and S data.

3. Materials and methods

A Berkovich indenter and two spherical tipped indenters (Synton, Switzerland) with nominal indenter radii of 5 and 20 μm were chosen for this study. Fused silica acted as the standard calibration material.

A healthy premolar tooth extracted for orthodontic reasons was embedded with a cold-curing epoxy resin (Epofix, Struers, Copenhagen, Denmark). The top surface and cross-sectional surface of the tooth were chosen for testing. The cutting directions were selected so that the enamel prism direction was perpendicular to the top surface and parallel to the cross-section. A low speed saw (Isomet, Buehler Ltd., Lake Bluff, USA) was used to cut the tooth under water irrigation. All the specimens were polished with two polishing machines (Grinder-Polisher, Buehler UK Ltd., Coventry, England; RotoPol-22, Struers, Copenhagen, Denmark) to 0.4 μm alumina polishing paste. Once prepared, the tooth was stored in distilled water at room temperature and the tests were conducted shortly after sample preparation. Previous investigations by Poolthong [32], using spherical tipped indenters over the same load range as investigated in this study, established the long-term stability of the hardness and elastic modulus of enamel stored at 4 $^\circ\text{C}$ in de-ionized distilled water.

The indentation experiments were performed using a nano-based indentation system (Ultra Micro-Indentation System, UMIS-2000, CSIRO, Australia). The surface-finished specimens were mounted on metal bases with wax using a paralleling machine (Leitz, Wetzlar, Germany). The mounting base contained a strong magnet to ensure an adequate contact with the test base in the UMIS. All the tests were done in distilled water.

The tooth samples, both top surface and cross-section, as well as fused silica were indented with a Berkovich tip and two spherical indenters

under a variety of loading forces from 1 to 450 mN. The distance between two separate indents was always greater than 40 μm to avoid the influence of residual stresses from adjacent impressions. The $H-\tan\theta$ curves were generated using the methods outlined above.

After indentation testing, scanning electron microscopy or SEM (XL-30, Philips, The Netherlands) was used to examine the residual indent impressions. Parts of the indented surface of the specimen were etched using 37% H_3PO_4 gel (Etchant gel, SDI Ltd., Australia) for 5 s to better highlight the surrounding damage. After etching, the samples were ultrasonic cleaned with pure ethanol for 10 s before gold coating. Secondary electron detector was selected for the SEM observation.

4. Results

4.1. Indenter tip calibration

The Berkovich indenter tip was calibrated with fused silica using the approach outlined by Oliver and Pharr [30]. The two spherical indenters were calibrated with fused silica using the methods described above to acquire the $a-h_p$ curve and its functional form. The results from more than 200 individual indentation tests on fused silica for each indenter were fitted to power-law functions as shown in Figs. 2a and b. The curve generated for the nominally 5 μm radius spherical tip (Fig. 2a) was fitted with the following expression:

$$a = 2.1818h_p^{0.5151} \quad (h_p \in [0, 1.3] \mu\text{m}, R^2 = 0.9975). \quad (9)$$

For the nominally 20 μm radius spherical indenter, two separate fitting functions were used because a single-fitting equation was not sufficiently accurate over the entire range. The curve fitting functions are

$$\begin{aligned} a &= 5.5006h_p^{0.5568} \quad (h_p \in [0, 0.2] \mu\text{m}, R^2 = 0.9956), \\ a &= 4.8486h_p^{0.4826} \quad (h_p \in [0.2, 0.7] \mu\text{m}, R^2 = 0.999). \end{aligned} \quad (10)$$

True tip radii, R , of both indenters at specific depths were calculated using Eq. (7). The initial contact radii of the tips were almost 5 and 20 μm , but thereafter the radii decreased with depth. The R^2 values associated with each equation indicate the quality of fit.

4.2. Sample load-displacement curves

Fig. 3 illustrates typical load-unloading curves of fused silica and enamel indented in this instance using the 20 μm indenter to a maximum force of 10 mN. Fused silica showed almost total elastic recovery upon unloading. In comparison with the top surface, the cross-section of enamel showed a greater inelastic response with the penetration depth almost 50% greater than the top surface.

4.3. Elastic modulus of enamel in both directions

Elastic modulus and hardness for the spherical tipped indenters at different contact depths in the following figures were calculated using Eq. (8) as well as curve fitting Eqs. (9) and (10). The Berkovich results were determined using the Oliver and Pharr [30] analysis. Eq. (2) with

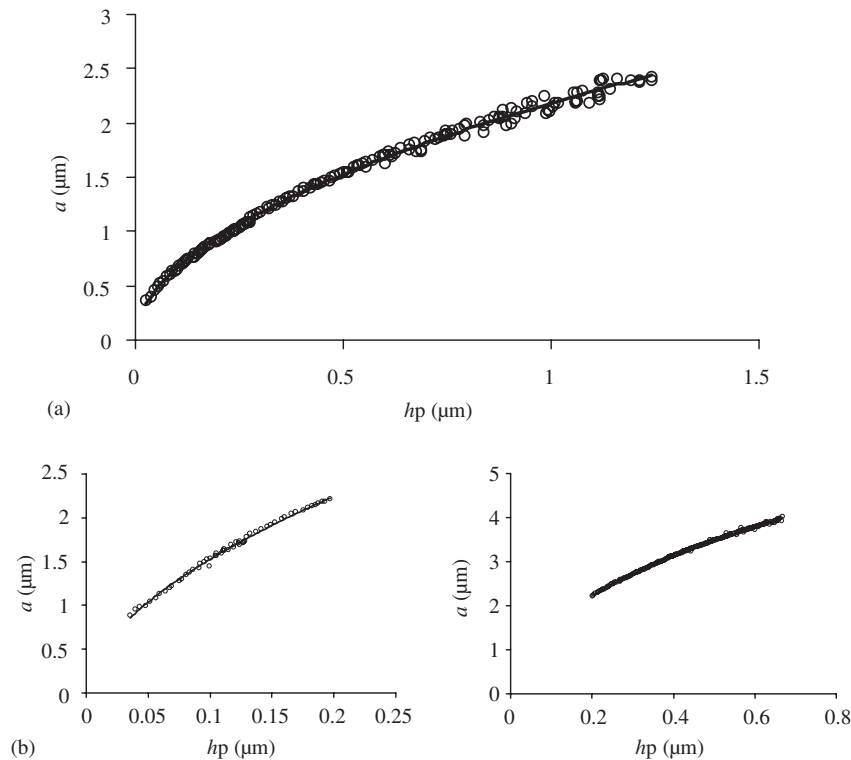


Fig. 2. Plots of contact radius versus contact depth of penetration for the (a) 5 μm and (b) 20 μm spherical indenter tips along with fitting equations.

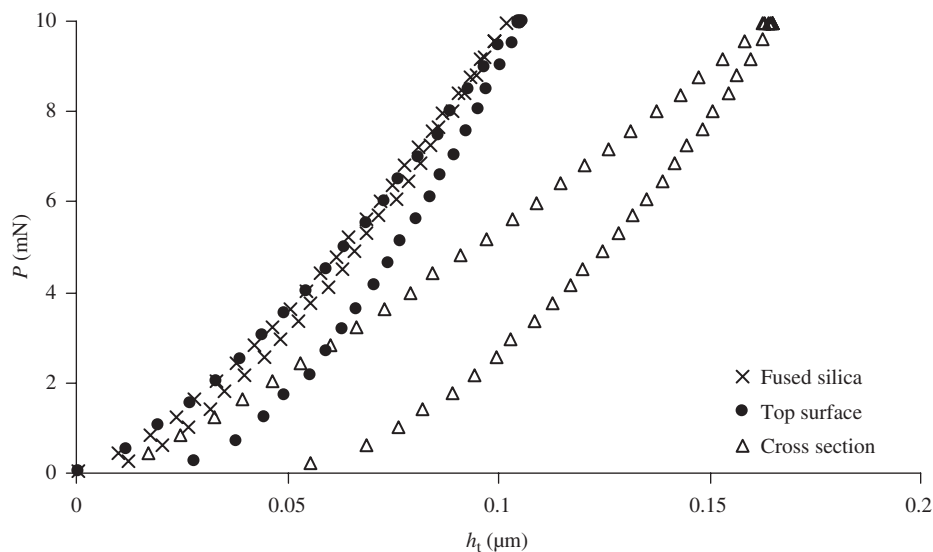


Fig. 3. Comparison of force-displacement curves for the 20 μm spherical indenter at 10 mN on fused silica as well as the top and cross-section surfaces of enamel.

known values of E and Poisson's ratio for diamond tip (1070 GPa and 0.07, respectively) along with a value of Poisson's ratio of enamel of 0.3 [33] was used to convert indentation modulus, E_{eff} , into E . Fig. 4 shows the variation of the elastic modulus and contact pressure versus indenter contact depth for indentations made on the top and cross-section surfaces. Unfortunately with the UMIS system it was difficult for us to control the exact

indent location. For this reason more than 100 indents on each surface were undertaken for each indenter.

The above figures show that analysis of the force-displacement data for different tips resulted in different elastic modulus and hardness values. Elastic modulus showed a trend to decrease with the increasing of contact depth. The variation of elastic modulus on the top surface was between 100 and 60 GPa, while this

value varies between 80 and 40 GPa on the cross-section surface.

In the case of hardness or contact pressure, H , the constant strain associated with the Berkovich tip produced a relatively stable hardness value, which was 5 ± 0.45 and 4.5 ± 0.45 GPa on top surface and cross-

section, respectively. In contrast, the hardness or contact pressure for the spherical tips increased with the increment of contact depth because of the increasing strain value. For both indenters, the contact pressure at the same depth of penetration was higher for the top surface.

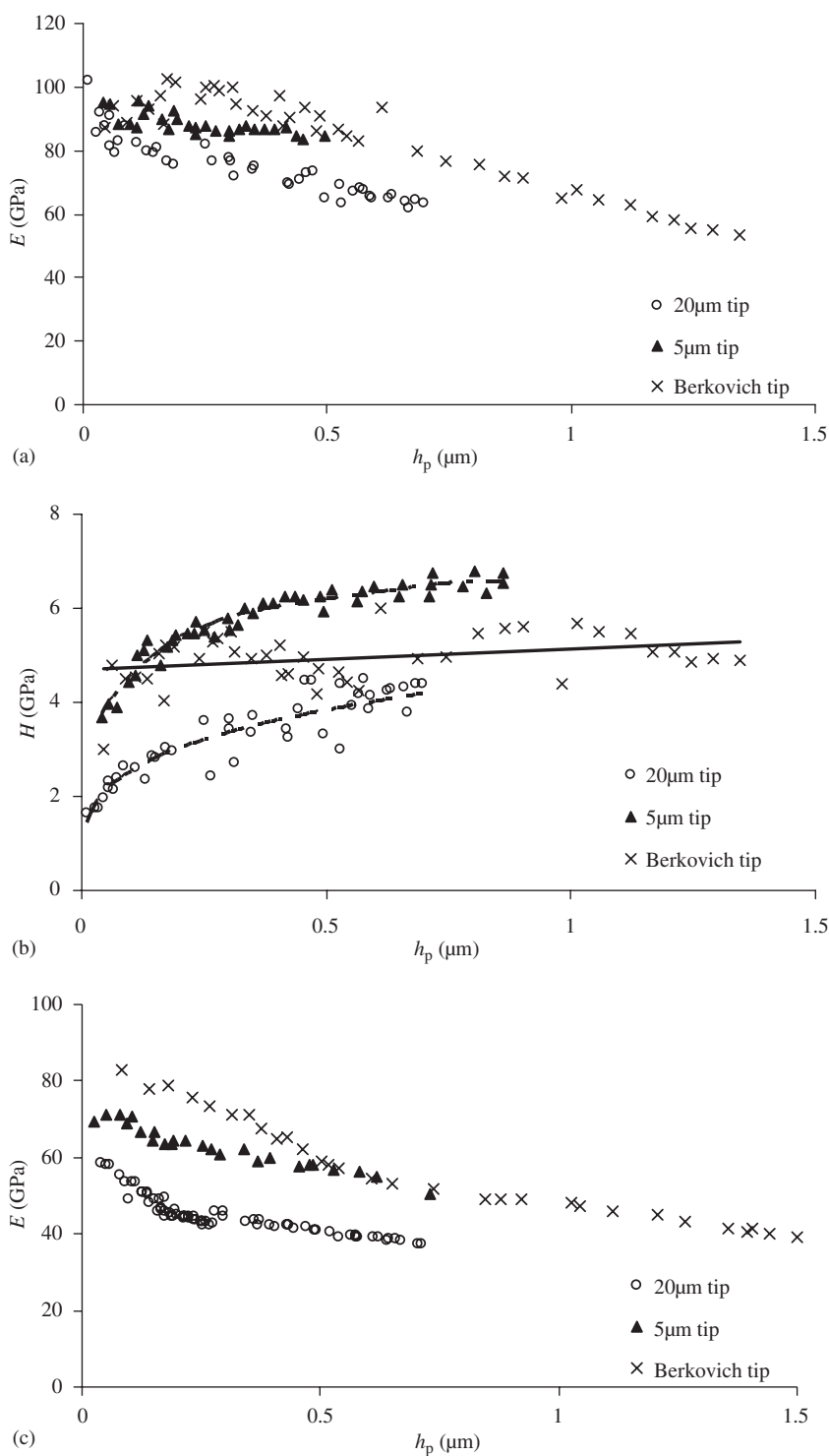


Fig. 4. Elastic modulus and hardness as a function of penetration depth of the (a), (b) top surface and (c), (d) cross-section of enamel obtained with Berkovich as well as the 5 and 20 μm radius indenters.

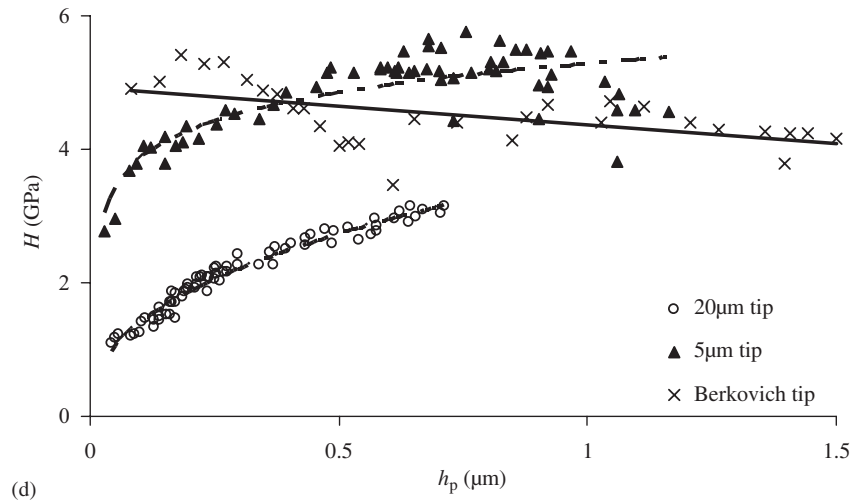


Fig. 4. (Continued)

4.4. Stress–strain response

Using Eqs. (5) and (6), the stress–strain response curves were obtained by including results for the 5 and 20 μm spherical tipped indenters for fused silica and the two directions of enamel and are plotted in Fig. 5. In these curves, each point is the result of one complete load–unload indentation test. In the case of the fused silica curve, the nominally 20 μm spherical indenter data only covered the purely elastic part of the stress–strain response while the 5 μm indenter extended into the elastic–plastic part. Although enamel has similar elastic modulus as fused silica, the stress–strain response is totally different. Even contact at the lowest loads investigated produced slight inelastic deformation on both enamel surface directions. Unlike fused silica, which had an elastic-to-plastic transition value of 10–12 GPa, no prominent transition point was found for enamel. In Figs. 5b and c, only data points whose $\tan \theta$ value was lower than 0.2 were selected from the 20 μm indenter. This point is taken up in more detail in the Discussion.

4.5. SEM observations

Fig. 6 summarizes the SEM images of 20 μm spherical indents on the surface of both directions of enamel specimen with images a and b from the top surface while c and d are from the cross-section surface. The maximum force for all indents in the figure was 350 mN. These observations show that human enamel has a typical prism diameter of approximately 4–5 μm and the thickness of organic-rich sheath of nearly 1 μm . Without etching, all indents looked smooth (see Figs. 6a and c) and the diameter of residual impression is nearly 10 μm . This value is very similar to that determined from analysis of the force–displacement data using the expressions given in Section 2.1. After being etched using 37% H_3PO_4 gel for 5 s, radial cracks were seen on the top surface (see Fig. 6b).

Most of these radial cracks occurred within the sheath area between the prisms. Ring like cracks and across prism cracks within, and at the periphery, of the contact area are observed on the etched surface of the indented cross-section (Fig. 6d).

5. Discussion

5.1. Spherical indentation techniques

The stresses and deflections arising from the contact between two elastic solids are of particular interest to those undertaking indentation testing. From Fig. 1, it is not difficult to see that $a-h_p$ curve describes the actual shape of the contact surface of the tip. Figs. 2a and b show the effective tip shape of the spherical indenters used in this study. It is not surprising that the indenter has a parabolic shape rather than perfect spherical form at this scale, because it is impossible to polish a smooth spherical shape accurately on a crystallographically anisotropic material such as diamond. The fitting equations also illustrate that the tips are not those of a standard sphere. However, Oliver and Pharr [22] have shown that their method may be used to calculate the contact pressure for any axis-symmetrical indenter tip, including these parabolic tips.

Because of the inevitable deviation of the tip shape from that of a sphere, we developed the present method to avoid using the parameter of tip radius, R , in the calculation. Therefore, this method is considered more practical and the use of very small spherical tips may become a more widely accepted technique for measuring the elastic modulus, hardness as well as stress–strain response of unknown materials.

5.2. Mechanical properties of enamel

Load–displacement curves (Fig. 3) illustrate that different surfaces have different mechanical response.

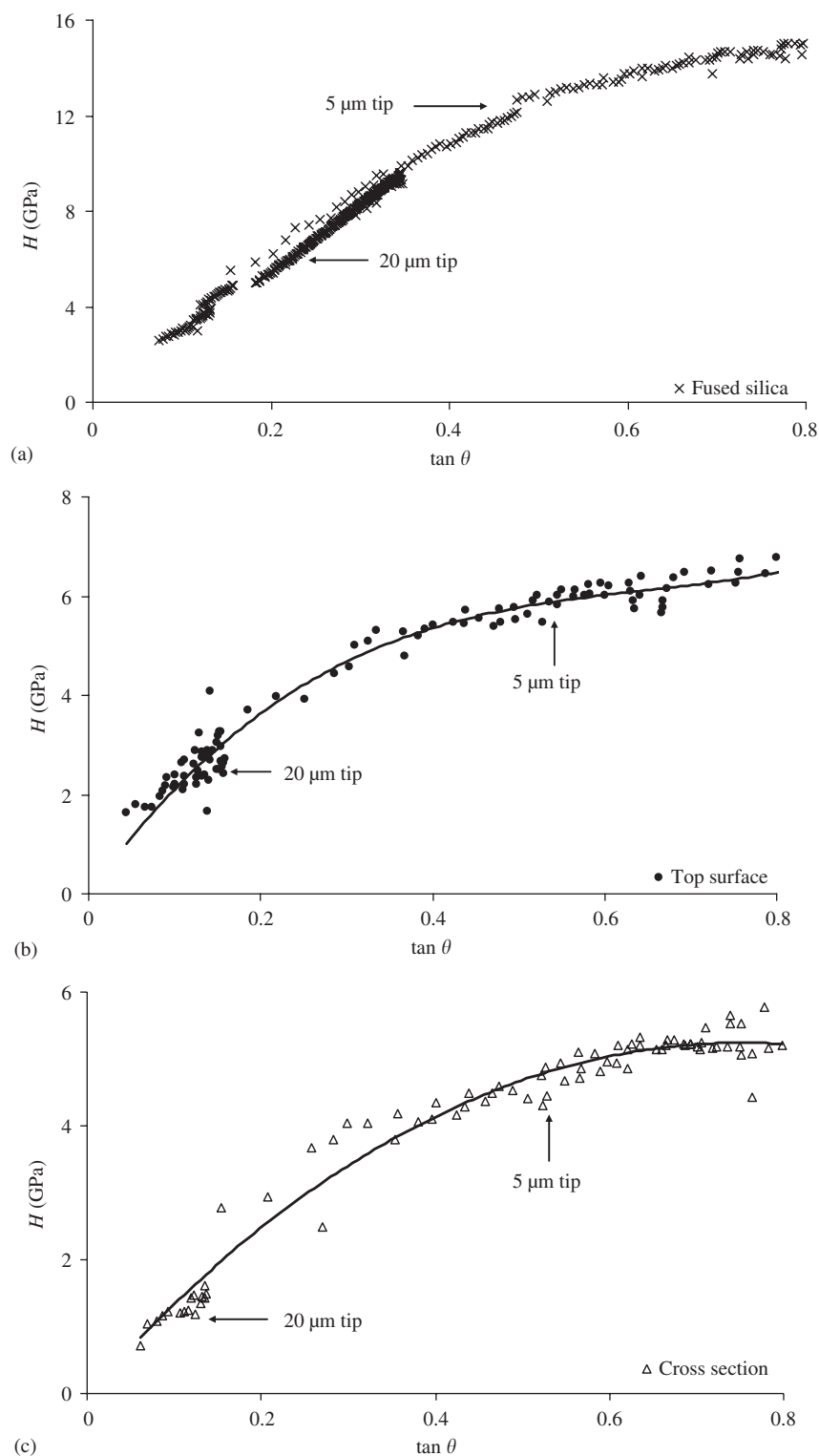


Fig. 5. Nano-indentation stress-strain curves on (a) fused silica, (b) top surface and (c) cross-section of enamel from two spherical tipped indenters.

Different tips also produced different elastic modulus at the same contact depth and both surfaces tested showed similar trends (see Fig. 4). This indicates that prism orientation and structure may play a role in determining the elastic modulus, hardness and stress-strain response.

5.2.1. Elastic modulus of enamel

Many tests have been done to measure the elastic modulus of enamel. Some of them along with the direction of testing and technique used are listed in Table 1.

Table 1 indicates that the value of the elastic modulus of enamel varies dramatically depending upon technique used

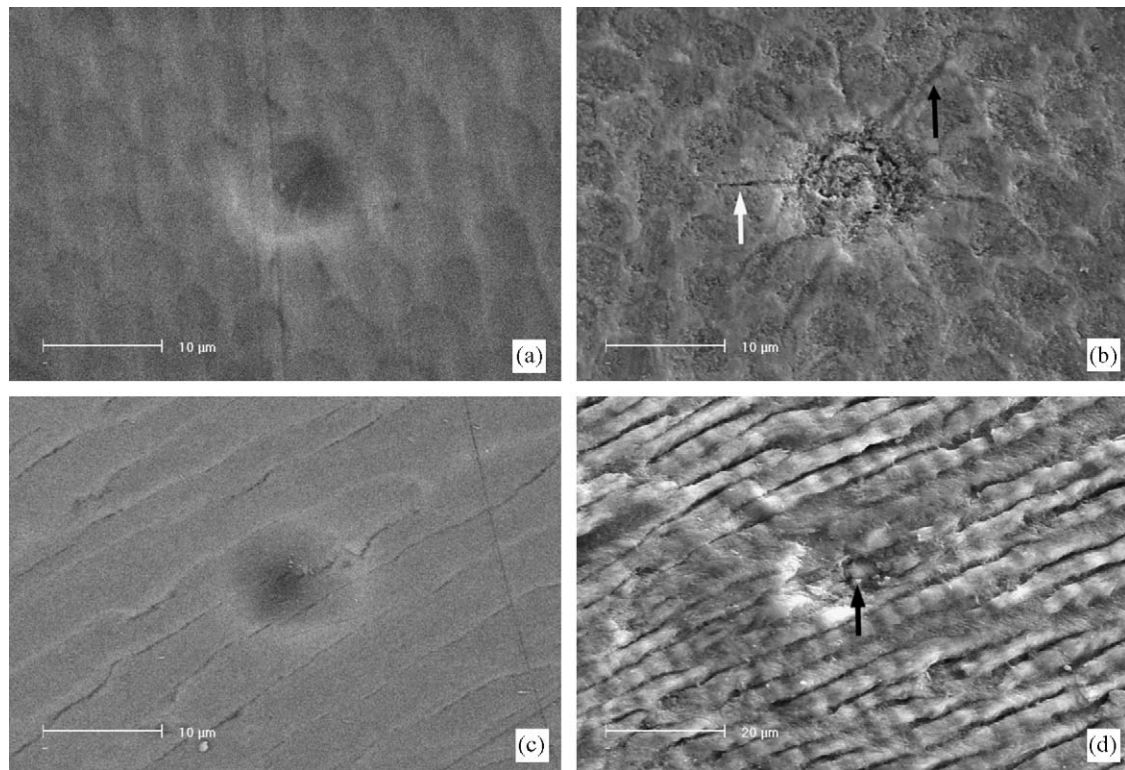


Fig. 6. SEM observations of 20 μm indenter residual impressions on enamel after loading to 350 mN. (a) is polished and (b) is etched top surface, while (c) is polished and (d) is etched cross-sectional surface of enamel. (Arrows indicate cracks and damage which are considered in the discussion.)

Table 1
Elastic modulus of enamel from different measurement methods

Author(s)	Prism orientation	Test method	Elastic modulus (GPa)
<i>Macroscale tests</i>			
Stanford et al. [18]	Cross-section	Compressive	32.4 ± 4.1
Craig et al. [11]	Top surface	Compressive	84.1 ± 6.2
Reich et al. [16]	Top surface	Acoustic impedance	76.5
Staines et al. [17]	Top surface	Spherical indentation	83
<i>Micro- and nano-scale tests</i>			
Xu et al. [20]	Top surface	Modified vickers	94 ± 5
	Cross-section	Indentation	80 ± 4
Willems et al. [19]	Top surface	Nanoindentation	90.59 ± 16.13
Cuy et al. [12]	Cross-section	Nanoindentation	47–120
Habelitz et al. [14]	Top surface	Nanoindentation	87.5 ± 2.2
	Cross-section	Nanoindentation	72.7 ± 4.5

and orientation investigated. One common point of agreement is that the elastic modulus of the top surface is always greater than the cross-section surface. The data generated in this study are within the range of results from other nano-indentation tests. We shall now consider a basis for the differences observed even within a single tooth over a small localized area as investigated in this study.

Enamel consists of two major structures, the prism or rod and surrounding organic-rich sheath structure. It has long been recognized that the mechanical properties of these two structures are different [5,9,33]. As a consequence the mechanical properties of enamel may be expected to

scale with the diameter of the prisms or rods, that is where the contact diameter is small compared with the rod diameter the elastic response would be dominated by the rod elastic properties whereas if the contact is large compared with the rod diameter then both the rods and sheath structure properties would play a role as modeled by Spears [9] for uniaxial loading. To illustrate this issue more clearly, the calculated elastic modulus values shown in Figs. 4a and c are replotted versus contact radius, a , for the three indenters as shown in Figs. 7a and b. In the case of the Berkovich indenter this was the effective contact radius determined from the contact area of the triangular

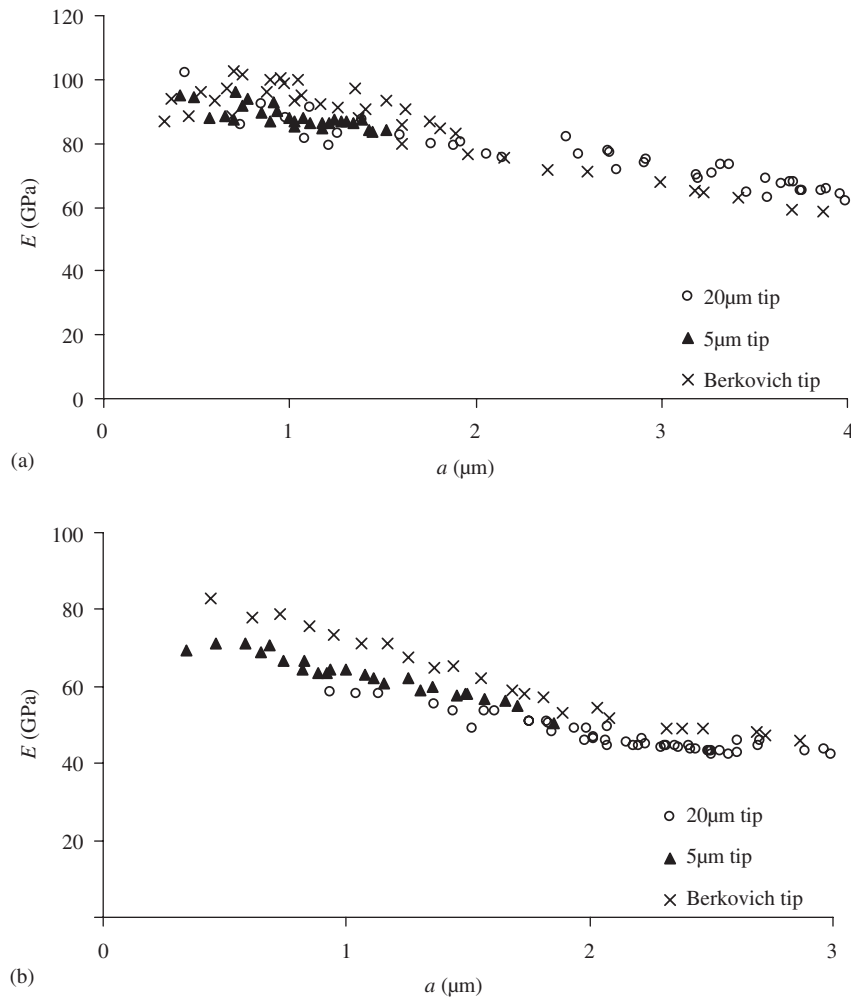


Fig. 7. Elastic modulus plotted against contact radius for the three different indenters on (a) the top surface and (b) cross-section surface of enamel.

impression. The resulting plots show a much more consistent decrease in E for all indenters in both directions. This illustrates that elastic modulus, in the case of enamel, is a function of contact area and the effective volume of the contact stress field from which the elastic modulus is determined, irrespective of what type of indenter tip is used.

The present observations support the very recent AFM-based nano-indentation results of enamel by Ge et al. [13], who reported that the nano-hardness and elastic modulus of sheaths were about 73.6% and 52.7% lower than those of the prisms. Such a lower elastic modulus of the organic-rich sheath will lower the modulus of the whole system when the contact area incorporates many rods. In an extreme condition, taking the diameter of prism as 5 μm , if the contact area is far smaller than this value and the contact point is on the central area of the prism, the organic sheath will play a minimal role and the elastic modulus will equal the elastic modulus of prism, which is composed of a high volume fraction of hydroxyapatite and has a high elastic modulus. With the increase of contact area, the stress field expands and the organic-rich sheath

will play a greater role in the resultant deflection of the contact area. Finally, the stiffness of the whole system decreases because of the lower elastic modulus of the sheath.

This prism-sheath unit related response results in the elastic modulus becoming a function of contact area and stress field. This may partially assist in explaining the wide range of elastic modulus reported in former results. For macro-scale tests such as compression tests, the contact area was much larger than the prism-sheath dimension and thus the elastic modulus of the whole prism-sheath unit system was measured. On the other hand, for nano-indentation tests, especially AFM-based tests, the contact area is much smaller than prism diameter. The results of this test reflect only the elastic modulus of prism which has higher volumetric fraction of hydroxyapatite. From Table 1, it is evident that macroscale tests generally had lower values of elastic modulus than micro-scale, especially nano-scale tests. In addition, the top surface values were always higher than cross-section values in accordance with Spears' FEA model [9].

5.2.2. Stress–strain response of enamel

The stress–strain curves in Fig. 5 for enamel includes all the 5 μm indenter data but only includes data for the 20 μm radius indenter where the contact diameter is less than the rod diameter. In this manner, a more consistent stress–strain response was found and as discussed above for the elastic modulus, the influence of the sheath structure is less dominant.

Although the difference of the elastic modulus between enamel and fused silica is small, the stress–strain response (Fig. 5) is totally different. This is mainly caused by the unique composite prism structure of enamel and its organic-rich sheath which makes the whole material more deformable than pure inorganic hydroxyapatite. The stress–strain curves, especially the inelastic part for the top surface, lie well above those for the cross-section surface. For indentations on the top surface, the force direction is parallel to the prism with the yield behavior primarily determined by shear deformation of the very thin protein “glue” between the individual apatite crystallites. In the case of the cross-section surface, the loaded prism deflects under the force because the contact load is supported not only by the prism structure but also by the more deformable and thicker sheath structure surrounding the prism parallel to the surface 3–5 μm below the surface.

5.3. Analysis of SEM photos

Fig. 6b shows the presence of radial cracks surrounding the contact impression. Two types of radial cracks may be distinguished in the picture. The first and most predominant types are inter-rod cracks growing along the sheath and propagating between prisms (black arrow in Fig. 6b). The second types are the inner-rod cracks that propagate within the prisms (white arrow in Fig. 6b). The first type of cracks indicates that fracture occurred around the enamel prisms rather than through them. This is in agreement with other tests on composite materials which showed that cracks tend to follow weak interfaces [34,35] as well as with previous observations on enamel by Rasmussen et al. [15] and White et al. [6]. The second form of cracks may have resulted from the acid etching procedure. After loading and unloading, residual stress accumulated in the enamel structure, especially in highly mineralized prism. During acid etching, the surface of the impression was damaged and the balance of residual stress changed and as a result a crack may propagate within the prism.

On the cross-section surface, the prisms and crystals direction are vertical to the loading force and act as strong integral units or fibers. Every prism is wrapped by organic sheath. The deformation in this direction is similar to the bending of beams. By comparing pictures c and d, it is clear that, after surface etching, substantial damage almost twice that of the surface impression is evident. In picture d, cracked prisms within the contact area had been exposed by acid and ring like cracks restricted in one prism are present (black arrow in Fig. 6d).

These preliminary observations of cracks in the indented surfaces indicate that the fracture mechanisms of enamel are influenced by the prism-sheath unit structure as also proposed by other researchers [6]. Further study is required to explore the exact fracture mechanisms of enamel especially inter prism fracture.

6. Conclusions

Nano-indentation is an attractive method for measuring the mechanical behavior of small specimen volumes. Using this technique, the mechanical properties of enamel were investigated in directions parallel and perpendicular to the prisms. From the present study using both pointed and spherical-tipped indenters the following conclusions are proposed.

1. Elastic modulus of enamel is influenced by prism-sheath structure and in the case of nano-indentation it is a function of contact area. This provides a basis along with the FEA model of Spears [9] to understand the different results reported in the literatures from macro- and micro-scale tests.
2. Anisotropic properties of enamel, which arise from the rod like prism-sheath structure, are well reflected in the stress–strain curves. The top surface is stiffer and has higher stress–strain response than an adjacent cross-section surface because of the greater influence of the prism sheaths in the latter behavior.

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